Mehrdad Mirzaei

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Summary

AI/ML Engineer with a Ph.D. in Computer Science and strong experience building and deploying production-grade Generative AI systems. Specialized in Retrieval-Augmented Generation (RAG), LLM-powered applications, and scalable ML pipelines on AWS (Bedrock, SageMaker) and Azure. Proven ability to design end-to-end AI solutions for natural language interfaces, automation, and decision support.

Professional Experience

Generative AI Engineer, *CGI Inc.* *(Client: Fannie Mae) - Reston, VA* *Feb 2025 – Present*

- Developed a Text-to-SQL chatbot on AWS SageMaker using Streamlit and AWS Bedrock, leveraging Retrieval-Augmented Generation (RAG) with Bedrock Knowledge Base to convert natural language into SQL, enabling non-technical users to efficiently query and interact with data through a scalable interface.
- Designed and implemented an agentic workflow using LangGraph to automate data extraction from PDF files, using LLMs to streamline and validate the loan application certification process.
- Fine-tuned large language models using LoRA on Azure Machine Learning (Azure ML) with question–answer datasets, improving the accuracy of the question-answering application by 20%.

Machine Learning Engineer, *ReadyOn AI* - *San Francisco, CA* *Jan 2024 – Jan 2025*

- Developed a scalable machine learning pipeline on Azure ML for predictive modeling, supporting workforce optimization and operational decision-making.
- Built data processing pipelines using Azure Databricks (Spark) to handle large-scale time-series data for feature engineering and model training.
- Designed and evaluated models using statistical analysis and experimentation (A/B testing, causal inference) to validate performance and business impact.

Data Scientist, *Epsilon* - *Chicago, IL* *April 2021 – Sept 2023*

- Developed and deployed machine learning models (Logistic Regression, XGBoost) for customer conversion prediction, improving targeting strategies and marketing effectiveness.
- Designed and implemented an end-to-end customer segmentation pipeline using clustering algorithms (K-means), enabling identification of customer groups and supporting targeted marketing strategies.
- Applied deep learning models (encoder–decoder architectures in PyTorch) for feature learning and dimensionality reduction on structured data.

Skills

- **Programming Languages**: Python, SQL, Java
- **Generative AI & LLMs:** GPT, Claude, LLaMA, RAG, LangChain, LangGraph, Prompt Engineering, Embeddings, Vector Databases, Agentic workflow
- **Machine Learning**: PyTorch, TensorFlow, Scikit-learn, Transformer Models
- **Cloud & AI Platforms**: AWS (Bedrock, SageMaker, S3), Azure (Azure ML, Data Factory), GCP
- **Data Engineering:** Spark (Databricks), Pandas, NumPy
- **MLOps & Deployment:** Docker, Kubernetes, MLflow, Airflow, CI/CD
- **Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB

Education

- SUNY at Albany, *Albany, NY* - ***Ph.D. Computer Science*** *Sept 2020*
- University of Tehran, *Tehran* - ***M.Sc. Software Engineering*** *Sept 2012*
- Shahid Beheshti University, *Tehran* - ***B.Sc. Software Engineering*** *Sept 2009*